

TOPIC 7: SANITATION AND WASTE MANAGEMENT

1. Meaning of Sanitation and Waste

Sanitation refers to the measures taken to protect public health by providing clean drinking water, proper disposal of human waste (sewage) and refuse, and maintaining general cleanliness of the environment.

Waste (or refuse) is any material that is no longer useful and is discarded. It includes solid waste (rubbish), liquid waste and gaseous waste.

2. Types and Sources of Waste

Types:

- **Solid waste** – plastic, paper, food scraps, metal, glass, polythene
- **Liquid waste** – sewage, wastewater from homes and industries
- **Gaseous waste** – smoke, exhaust fumes, industrial gases
- **Biodegradable waste** – can be broken down by microorganisms (food, leaves, paper)
- **Non-biodegradable waste** – cannot be easily broken down (plastics, glass, metals)

Sources: homes, schools, markets, industries, farms, hospitals, streets.

3. Waste Disposal Methods

- **Open dumping** – throwing waste on open ground (unhygienic and illegal in many places)
- **Landfill** – burying waste in designated sites
- **Incineration** – burning waste under controlled conditions
- **Composting** – converting organic waste into manure
- **Recycling** – converting waste materials into new products
- **Sewage treatment** – treating liquid waste before release into the environment

4. Recycling and Reusing Waste Materials

Recycling means processing used materials so they can be used again (e.g. plastic bottles → new plastic products, paper → new paper, metal cans → new metal items).

Reusing means using an item again for the same or a different purpose without major processing (e.g. using a jam jar as a container, using old clothes as cleaning rags).

Benefits: reduces the amount of waste, saves raw materials and energy, reduces pollution, creates jobs.

5. Good Sanitation Practices

- Use toilets properly and keep them clean.
- Dispose of refuse in covered bins and take it to approved collection points.
- Do not dump waste in drains, rivers or open spaces.
- Wash hands regularly with soap.
- Keep the home, school and surroundings clean.
- Separate biodegradable and non-biodegradable waste where possible.

6. Summary

Sanitation protects health through cleanliness and proper waste disposal. Waste can be solid, liquid or gaseous; biodegradable or non-biodegradable. Proper disposal methods, recycling and reusing reduce environmental pollution and promote a healthier community.

— End of JSS 1 Intermediate Science First Term Notes —